

Government Shutdowns and Food Access Inequity: Identifying Communities Most at Risk

DATAJAM Project

MARCOS LOPEZ AND SAMUEL EUN



OUR PROPOSAL

- What we wanted to find:
 - Do low-income communities that rely on SNAP benefits face greater risk of food insecurity
- Relevance and Importance:
 - With recent news of SNAP funding cuts due to government shutdown, we wanted to find out if people were at risk due to delays and interruptions of federal food assistance
- Hypothesis:
 - Counties with high SNAP participation, lower median income, and limited access to grocery stores will exhibit significantly higher vulnerability to food insecurity during a government shutdown. These communities are more dependent on federal support and already face structural barriers to food access



DATA ACQUISTION

- To analyze how the recent government shutdown may disproportionately impact communities with limited food access and high reliance on federal assistance, our project uses publicly available datasets from three federal sources. Each dataset provides variables needed to measure income, food access, and SNAP dependence at the county level.
- Data is from the USDA Food Access Research Atlas, the USDA SNAP participation dataset, and the 2019 ACS 5-year estimates API.
- Citations:
 - <https://www.ers.usda.gov/data-products/food-access-research-atlas/download-the-data>
 - <https://www.fns.usda.gov/pd/supplemental-nutrition-assistance-program-snap>
 - <https://www.census.gov/data/developers/data-sets/acs-5year/2019.html#list-tab-1806015614>

FINAL DATASET

- Stripped all of our datasets to get these columns
 - Aggregated tract level food atlas data to county level
 - Extracted fips for all data
 - Merged on fips and saved as csv for analysis
-
- Link to download dataset:
 - o [github](#)

Column name	Description	Source
Fips	Federal Information Processing Standards county code (unique county identifier)	All datasets (merge key)
State	U.S. state name	All datasets
County	County name	All datasets
Urban	Proportion of county population living in urban census tracts	USDA Food Access Research Atlas
LILATracts_1And10	Share of low-income population living more than 1 mile (urban) or 10 miles (rural) from a supermarket (core <i>food desert</i> indicator)	USDA Food Access Research Atlas
HUNVFlag	Share of households with household has uno vehicle + no vehicle & low access to grocery stores (transportation + access barrier)	USDA Food Access Research Atlas
SNAP_Participants	Number of individuals receiving SNAP benefits in the county	SNAP County Level and Participation Data
MedianIncome	County median household income	ACS 5-Year Estimates
PovertyPercent	Percent of county population living below the federal poverty line	ACS 5-Year Estimates
TotalPopulation	Total population of the county	ACS 5-Year Estimates



Income Group	SnapRate	FoodAccessScore	PovertyPercent	HardshipIndex
Low Income	0.189	0.64	21.487	24.033
Middle Income	0.115	0.379	14.057	15.986
High Income	0.07	0.256	9.929	11.431

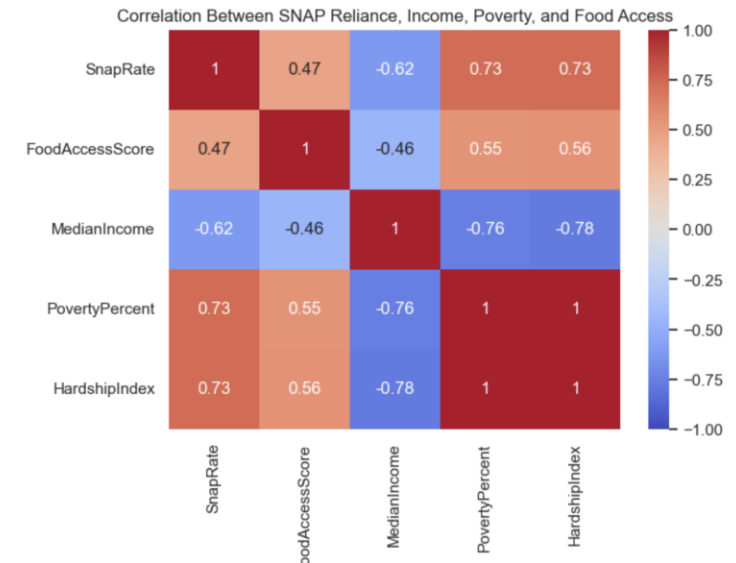
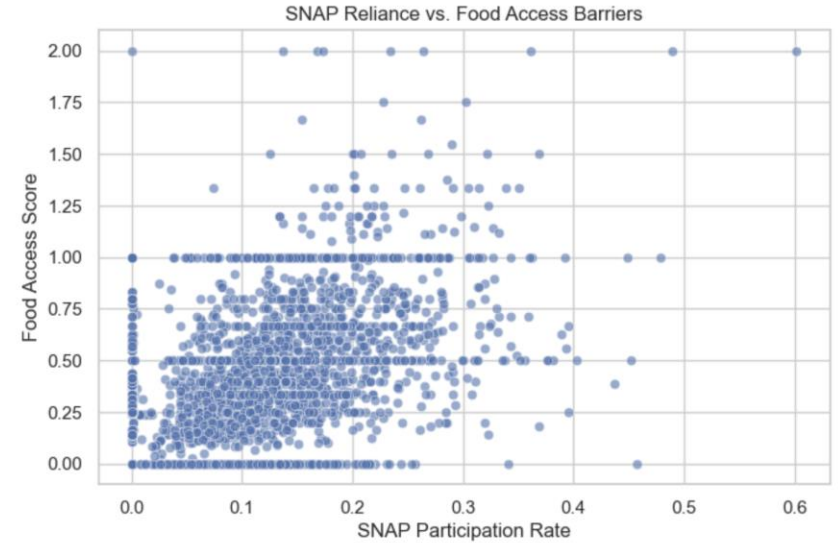
ANALYSIS

Income Group	MedianIncome	FoodAccessScore	PovertyPercent	HardshipIndex
Low SNAP reliance	64171.067	0.252	10.361	11.994
Medium SNAP reliance	53502.678	0.397	14.19	16.108
High SNAP reliance	42837.913	0.626	20.918	23.343

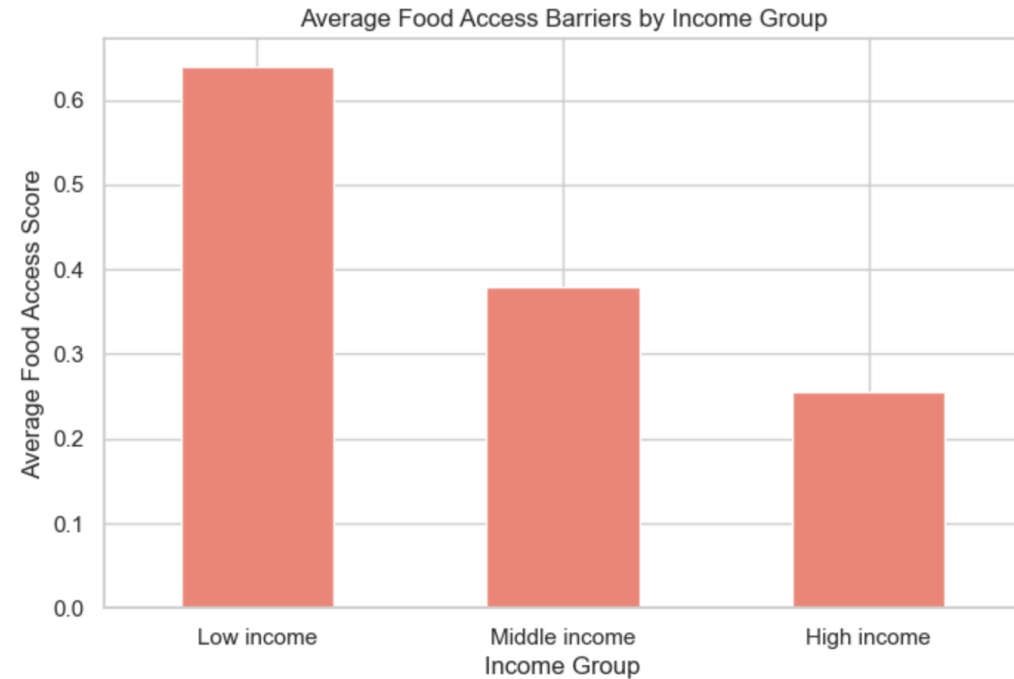
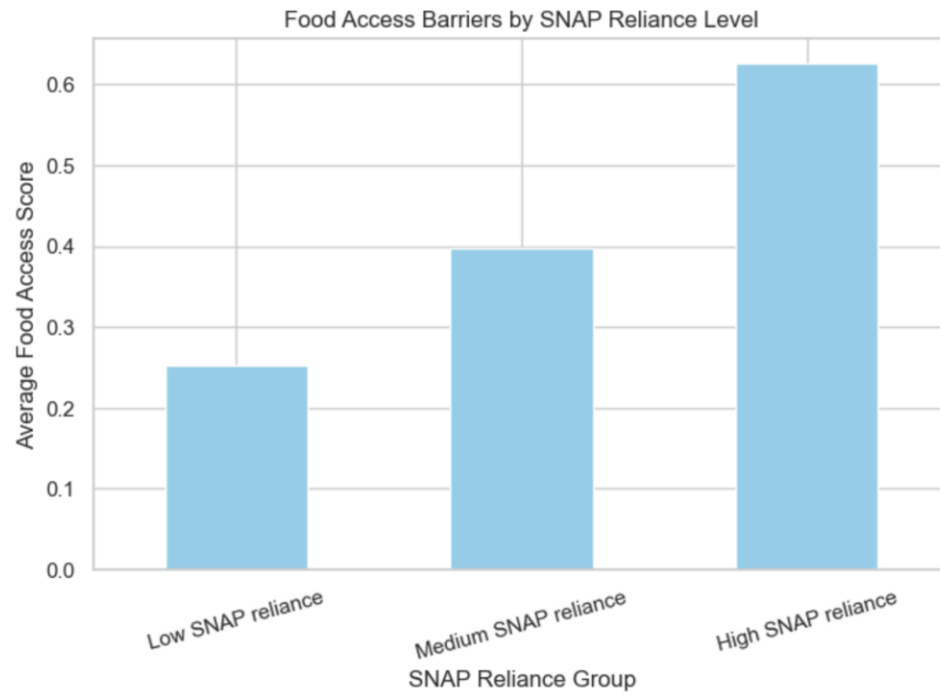
- Derived 3 features for analysis :
 - SNAP participation rate (SNAP participants / Total Population)
 - $df["SNAP_Participants"] / df["TotalPopulation"]$
 - Food Access Score (food desert + no vehicle/low access), low is more access
 - $df["LILATracts_1And10"] + df["HUNVFlag"]$
 - Hardship Index (poverty + low income)
 - $df["PovertyPercent"] + (1 / df["MedianIncome"] * 100000)$
- Group Counties by Income and SNAP reliance
 - Low income, Middle income, High income (based on Median Income)
 - Low SNAP reliance, Medium SNAP reliance, High SNAP reliance (based on SnapRate)
 - Both groupings were split into equal thirds using qcut in pandas
- Made Summary tables for both with means

MODELS AND VISUALIZATIONS

- Created the following models and Visualizations
 - Correlation Matrix (shown as heatmap)
 - ScatterPlot (SNAPRate vs Food Access)
 - Barchart (Income vs Avg Food Access)
 - Barchart (SNAP reliance vs Avg Food Access)



CONTINUED VISUALS



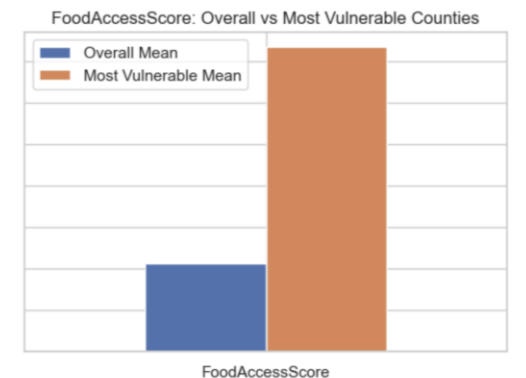
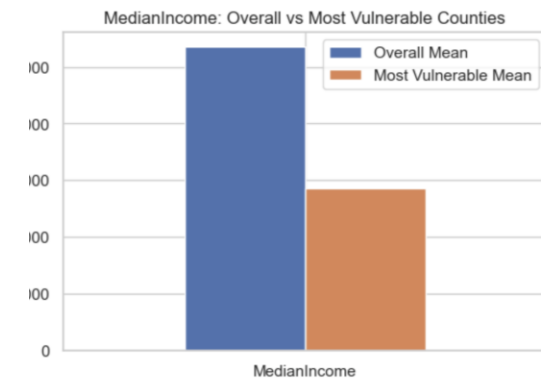
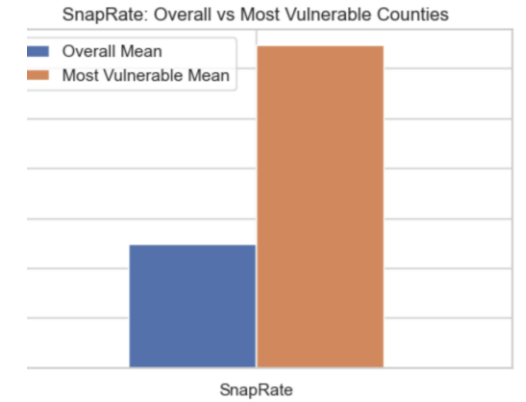
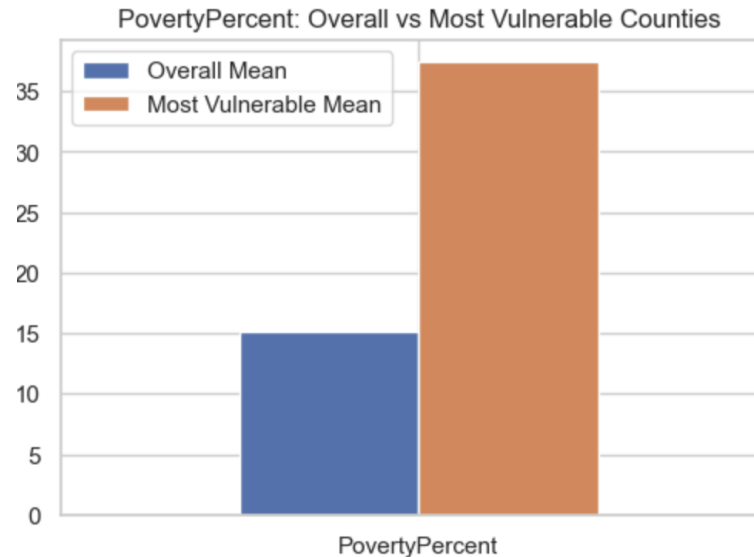
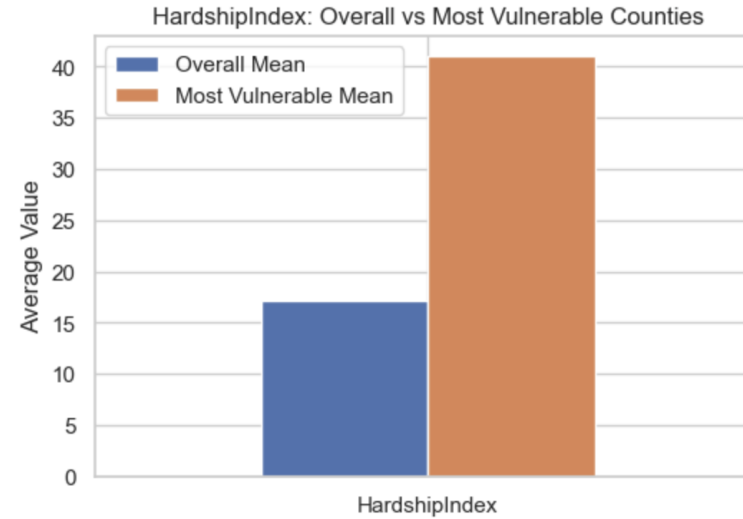
Most Vulnerable

	County	State	SnapRate	FoodAccessScore	MedianIncome	PovertyPercent	VulnerabilityScore
1028	Todd County	South Dakota	0.489259	2.000000	24331	55.5	2.997904
592	East Carroll Parish	Louisiana	0.350761	1.333333	22346	44.6	2.976111
1325	Clay County	Kentucky	0.338963	1.333333	26840	36.6	2.967728
2224	Leflore County	Mississippi	0.322839	1.250000	26735	36.8	2.959137
2226	Claiborne County	Mississippi	0.304104	1.333333	29338	41.0	2.955993
768	Randolph County	Georgia	0.321526	1.500000	31699	32.9	2.954946
2150	Zavala County	Texas	0.301935	1.750000	34459	33.8	2.949916
558	Leslie County	Kentucky	0.290188	1.333333	31627	38.0	2.948868
2194	Holmes County	Mississippi	0.298585	1.200000	21504	42.4	2.947611
1479	Luna County	New Mexico	0.368683	1.500000	29360	27.7	2.945306
1484	Noxubee County	Mississippi	0.314673	1.333333	33784	32.5	2.945096
147	Bolivar County	Mississippi	0.285795	1.375000	29854	35.0	2.944258
2129	McKinley County	New Mexico	0.332049	1.117647	33834	34.8	2.943001
2203	Quitman County	Mississippi	0.261862	1.666667	25283	35.6	2.930637
2085	Dimmit County	Texas	0.264131	2.000000	27161	33.7	2.929170

- Created a vulnerability score to located the 15 most vulnerable counties
 - `df["VulnerabilityScore"] = (`
`df["SnapRate"].rank(pct=True) +`
`df["FoodAccessScore"].rank(pct=True)`
`) +`
`df["HardshipIndex"].rank(pct=True)`
`)`

Were we correct ?

- Using our most vulnerable counties and our overall dataframes, we found the means of our engineered variables and compared





FINAL VERDICT

Our Hypothesis was correct! We believe that the evidence strongly supports our claim that low-income communities that rely on SNAP benefits face greater risk of food insecurity. Vulnerable communities on average had much higher SNAP participation, worse food access barriers, MUCH lower incomes, more than double the poverty, and nearly 3x the hardship than the average community overall. These findings validate our hypothesis and emphasize the urgent need to protect vulnerable counties from interruptions in federal assistance.